



NPTEL Online Certification Course
Generative AI for Computer Vision, IIT Guwahati
Assignment 5: Detailed Solution
Indian Institute of Technology Guwahati



1. What is the main purpose of Backpropagation Through Time (BPTT) in an RNN?

- a) To generate images
- b) To train the RNN by computing gradients over time (✓)
- c) To increase image resolution
- d) To remove the hidden state

Answer: b

Detailed Explanation: BPTT is used to train RNNs by propagating errors backward through the different time steps of the sequence.

2. In a character-level RNN, the model processes:

- a) Only numerical tables
- b) Only complete images
- c) Individual characters in a sequence (✓)
- d) Only audio signals

Answer: c

Detailed Explanation: In character-level modelling, an RNN processes a sequence one character at a time and can learn patterns in the sequence.



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3. Which problem is commonly associated with traditional RNNs?

- a) Vanishing gradient problem (✓)
- b) Image resizing problem
- c) Object detection problem
- d) Image compression problem

Answer: a

Detailed Explanation: During training, gradients in traditional RNNs can become very small as they are propagated through many time steps, making it difficult to learn long-term dependencies.

4. Which neural network architecture is specifically designed to handle long-term dependencies in sequential data?

- a) CNN
- b) LSTM (✓)
- c) SVM
- d) K-Means

Answer: b

Detailed Explanation: LSTM networks use special gates and a memory cell to preserve important information over longer sequences.



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5. What is the main purpose of the attention mechanism in sequence-to-sequence models?

- a) To resize images
- b) To remove all input information
- c) To reduce the number of classes
- d) To focus on the most relevant parts of the input sequence (✓)

Answer: d

Detailed Explanation: Attention allows the model to give more importance to relevant parts of the input when producing an output.

6. Computer vision mainly deals with:

- a) Understanding and analyzing images and videos (✓)
- b) Managing computer networks
- c) Designing databases
- d) Writing operating systems

Answer: a

Detailed Explanation: Computer vision enables computers to extract useful information and understand visual data such as images and videos.



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7. Which of the following is a common computer vision task?

- a) Image classification
- b) Object detection
- c) Semantic segmentation
- d) All of the above (✓)

Answer: d

Detailed Explanation: Image classification, object detection, and semantic segmentation are important applications of computer vision.

8. What is image super-resolution used for?

- a) Increasing the resolution or quality of an image (✓)
- b) Detecting passwords
- c) Removing the image completely
- d) Classifying text documents

Answer: a

Detailed Explanation: Image super-resolution aims to generate a higher-resolution image from a lower-resolution input.



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9. Which of the following is an application of Generative AI in computer vision?

- a) Image-to-image translation
- b) Medical image synthesis
- c) Face synthesis
- d) All of the above (✓)

Answer: d

Detailed Explanation: Generative AI can create or transform visual content for applications such as image-to-image translation, medical image synthesis, and face synthesis.

10. What is image in-painting?

- a) Converting an image into a table
- b) Reducing the number of image classes
- c) Filling or reconstructing missing regions of an image (✓)
- d) Removing all colors from an image

Answer: c

Detailed Explanation: Image in-painting uses surrounding visual information to reconstruct missing, damaged, or unwanted regions of an image.